LESSON PLAN

HIGH SPEED FLIGHT (12)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner
5. Specific Objectives	12.1.2.1 Trainee will be able to explain propellers and its associated maintenance practice

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propeller construction diagrams; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft propeller?
(b) Name two types of propellers used in aviation.
(c) How does a propeller generate thrust?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General construction of Dornier propeller • Blade construction materials and structure • Hub assembly and attachment points • Blade pitch change mechanism components • Hydraulic pitch change system operation	L & Ds	PPT, IB	15 min
(b)	• Spinner construction and aerodynamic function • Maintenance practices for propeller blades • Inspection criteria for pitch change mechanism • Common defects in propeller construction • Safety precautions during propeller inspection	L & Ds	PPT, PDF	20 min
(c)	• Summary of propeller construction details • Key maintenance points for pitch change • Review of blade structural integrity checks • Importance of spinner alignment and fit • Preparation for ground run checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the pitch change mechanism in variable pitch propellers.
(b) Describe the primary maintenance checks required for propeller blades.
(c) Why is the spinner critical for propeller aerodynamic efficiency?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between fixed pitch and variable pitch propellers, referencing the construction details covered in this lesson to prepare for LP 06 ground run checks.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various checks on propeller on the ground during ground run. General precautions to be observed while working on propeller, ground run	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min